



Preliminary Thermal-Hydraulics and Neutronics Studies on HEXANA Pool-Type Sodium-cooled Fast Reactor Concept

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ABSTRACT

HEXANA aims to develop an Advanced Modular Reactor (AMR) of the Sodium-cooled Fast Reactor (SFR) type to deliver flexible, low-carbon heat and power for energy-intensive industrial applications. This paper presents the key features of the proposed reactor concept, with emphasis on the primary circuit layout, core design, and challenges arising from the compact configuration.

The two primary pumps are positioned asymmetrically within the reactor vessel and a detailed CFD analysis of the lower plenum was therefore conducted to evaluate flow distribution at the core inlet. Results indicate a nearly uniform flow, with standard deviations below 0.04 kg/s. In parallel, a system-level thermal-hydraulic model of the HEXANA reactor was developed in the Dymola environment. Dynamic simulations of representative transients, including unprotected loss-of-flow (ULOF) and loss-of-heat-sink (ULOHS) scenarios, were performed to provide a preliminary assessment of the reactor response under abnormal conditions.

The compact design also introduces significant spatial constraints for the neutronic core layout. Preliminary neutronic studies, conducted using an in-house tool based on OpenMC, indicate that the proposed configuration is viable within these constraints. Overall, the presented analyses support the feasibility of the HEXANA concept at a preliminary design level. Further work is required to refine the core and primary circuit designs and to confirm these initial findings through more detailed analyses.

Keywords: Sodium-cooled Fast Reactor, Thermal-Hydraulic Analysis, Core Neutronics Analysis

1. INTRODUCTION

The achievement of long-term climate and energy policy objectives requires energy technologies capable of delivering large amounts of low-carbon energy in a reliable and sustainable manner. Beyond electricity generation, nuclear technologies are increasingly considered for supplying heat and other energy services needed to decarbonize industrial sectors that are difficult to electrify.

In this context, Advanced Modular Reactor (AMR) systems, encompassing Generation III/III+ and Generation IV designs, are being investigated as complements to existing large-scale nuclear power plants to address evolving energy market needs [1]. These concepts aim to extend nuclear applications through higher operating temperatures, improved thermal efficiency, and optimized fuel cycles, including fast-neutron systems that enable more efficient use of nuclear materials. In addition, low-power reactors (<300 MWe) offer potential advantages such as modular construction, phased deployment, and improved adaptability to specific grid or industrial demands. The introduction of advanced features and compact configurations, however, also raises significant technical challenges related to reactor core design, thermal-hydraulic behavior, safety demonstration, and system integration. Addressing these challenges through rigorous analysis and progressive design development is therefore a prerequisite for the future deployment of advanced reactor systems.

The motivation for AMR deployment is particularly strong in Europe, where energy-intensive industrial processes—including chemical production, petrochemicals, e-fuel synthesis, and steelmaking—require large and continuous energy inputs, with approximately 90% currently supplied by fossil fuels [2]. These sectors account for about 9 GtCO₂ emissions annually at the global level [3], and their decarbonization is a key objective of European climate policy [4]. Moreover, many industrial processes require energy sources that are flexible and readily controllable; in this context, AMRs represent a potential low-carbon alternative to fossil fuels while avoiding some inefficiencies associated with extensive electrification.

This paper presents the HEXANA reactor concept, which is part of an integrated energy platform designed to decarbonize energy-intensive industrial sectors such as chemical, petrochemical, and steel production. Sections 2 and 3 describe the core layout, preliminary neutronic studies at the Beginning of Life (BOL), and thermal-hydraulic analyses conducted at both the local scale and the system level. Finally, Section 4 provides a few concluding remarks.

2. HEXANA REACTOR CONCEPT

The HEXANA reactor concept is an AMR of the Sodium-cooled Fast Reactor (SFR) type, developed as an integrated energy system for both industrial heat and electricity production. The platform comprises a pair of AMRs, each with a thermal power of approximately 440 MW_{th}, capable of supplying electricity and process heat across multiple temperature levels, with a minimum outlet temperature of 465°C. The concept includes a Thermal Energy Storage (TES) system, an intermediate molten salt loop with two reservoirs, which provides additional operational flexibility and allows full thermal decoupling between the nuclear modules and the energy conversion system.

2.1. Primary Circuit Characteristics

Each reactor features a pool-type SFR configuration, so that the entire primary coolant system is submerged in a pool of liquid sodium. To optimize compactness, both intermediate heat exchangers and primary pumps are arranged asymmetrically around the core, as shown in Figure 1.

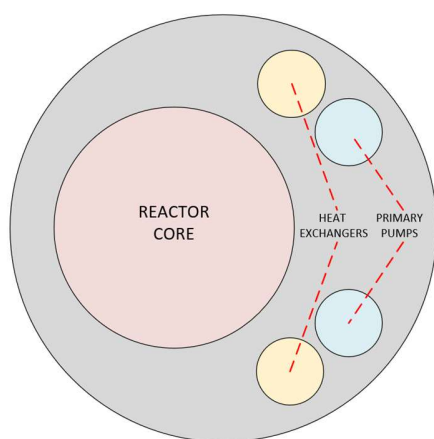


Figure 1. HEXANA primary circuit layout.

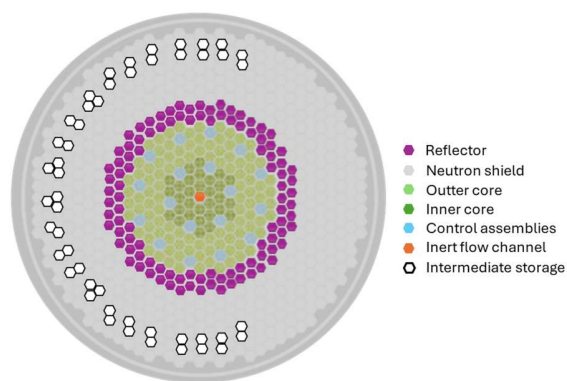


Figure 2. HEXANA core layout.

Due to the compact design of the HEXANA reactor, the space allocated to the reactor core is also limited, leading to highly constrained design choices. The outer diameter of the vessel that accommodates the core is 4.2 m. Within this volume must fit the fissile zone that generates nuclear power, the reflectors

(ensuring neutron economy), and the lateral neutron shielding structures protecting both the internal structures and the vessel.

Figure 2 illustrates the layout of the core components. The core includes 132 fuel assemblies, 102 reflector assemblies made of steel and arranged in two concentric rings around the fissile zone, and the borated steel neutron shield that fills the remaining space. The core is controlled by 18 control assemblies, which are used both to compensate for reactivity losses, to reach a safe state in case of accident and to flatten the power distribution, thereby ensuring a uniform power profile and avoiding local hot spots. Based on the experience of previous reactors such as Phénix and Superphénix, the core also includes a central inert flow channel designed to mitigate severe accident scenarios. This channel is filled with stagnant sodium and directly connected to the lower plenum. No detailed study has yet been conducted to assess the adequacy of this channel. Future work will focus on further clarifying this aspect. Nevertheless, its location is regarded as the most appropriate for managing the flow, based on operational experience and feedback. In addition, the white hexagons shown in Figure 2 correspond to intermediate storage locations used to allow the residual power of the assemblies to decay before they are fully unloaded and transferred to the washing station.

Table I. Main design parameters of the HEXANA core.

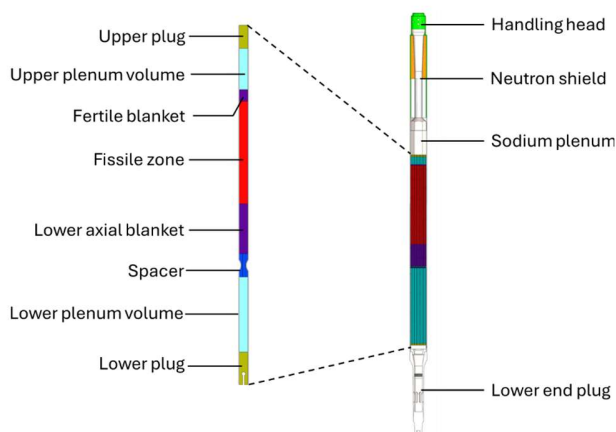


Figure 3. Fuel pin (left) and fuel assembly (right).

Thermal power	440 MWth
Coolant	Liquid sodium
Core configuration	Hexagonal lattice
Fissile core height	<100 cm
Total height	<500 cm
Upper sodium plenum	30 cm
Core outer diameter	420 cm
Number of fuel assemblies	132
Number of control rods	18
Pins per assembly	169
Fuel enrichment: MOX	2 zones
Mechanical spacing	Helicoidal wire (0.1 cm)

The design of the fissile zone results from a set of requirements aimed at achieving improved performance in terms of cycle length, power availability, and safety, and keeps the concept within the French standards. The fuel is MOX (Mixed Oxide of plutonium and uranium) based on a depleted uranium matrix. It is divided into two plutonium enrichment zones to flatten the neutron flux distribution. The average plutonium content is between 15% and 30% dependent on the area, which ensures compatibility with existing fuel fabrication capabilities and remains below the plutonium solubility limit. The fuel is arranged in hexagonal assemblies enclosed by a hexagonal steel wrapper. Each assembly contains 169 fuel pins arranged in a triangular lattice. Each pin includes a fissile MOX section (80 cm high), plenum volumes to accommodate fission gas release, and fertile blankets. The cladding material is 15-15Ti austenitic stainless steel, selected for its high resistance to irradiation and its proven operational performance under fast neutron conditions. Figure 3 presents the components

of a fuel pin and a HEXANA fuel assembly. The main characteristics of the HEXANA core are summarized in Table I. The presence of an upper sodium plenum can be observed, whose function is to promote the most negative possible voiding effect and to improve the transient behavior of the core.

3. PRELIMINARY STUDIES ON THE HEXANA REACTOR

The HEXANA project is currently in the conceptual design phase, and several studies have already been carried out in the field of core analysis with the aim of supporting reactor design development and assessing critical design aspects. These studies, briefly described in this section, mainly focus on neutronic and thermal-hydraulic analyses.

3.1. Lower Plenum CFD Analysis

Due to the compact design of the reactor vessel, the thermal-hydraulic configuration of the HEXANA reactor is inherently asymmetric, which may lead to non-uniform cooling of the core — a phenomenon that must be minimized to ensure safe and reliable operation. It is therefore essential, even at the preliminary design stage, to assess whether the current configuration is acceptable or if geometric modifications are required. As a preliminary geometry adjustment to mitigate uneven flow distribution among the assemblies, the geometry of the lower plenum has been modified by introducing an internal shell surrounding the bundle inlet region. The lower plenum is a key structural component of the reactor, designed to collect the pressurized cold sodium exiting the primary pumps and redistribute it to the various assemblies. With the introduction of the inner shell, a radial core coolant supply has somehow been recreated. To test the proposed solution and evaluate its effectiveness, a detailed Computational Fluid Dynamics (CFD) model of the lower plenum was developed by means of *Star-CCM+* software [5]. Main mesh nodal characteristics are reported in Table II.

Table II. Mesh nodal characteristics.

Number of cells	30 M
Average cell quality	0.88
Average Aspect Ratio	0.67
Average skewness angle	23.3

Steady-state and isothermal simulations were performed using a realizable $k-\varepsilon$ turbulence model for this preliminary work, and porous regions have been introduced to regulate the mass flow rate in reflectors and neutron shielding regions according to design specifications.

Streamlines in Figure 4 show qualitatively how the introduction of an inner shell helps recreating a core radial coolant supply. Such flow is also homogeneous, with no major asymmetries found, as reported in Figure 5, where the difference between fuel assembly flow rate and the core average value is shown. A sensitivity study was also performed to assess the flow redistribution within the core under asymmetric operation of the primary pumps, considering as an example the complete shutdown of one of the two pumps. The resulting flow distributions were compared by calculating the standard deviation on each core ring. For both cases results are reported in Table III.

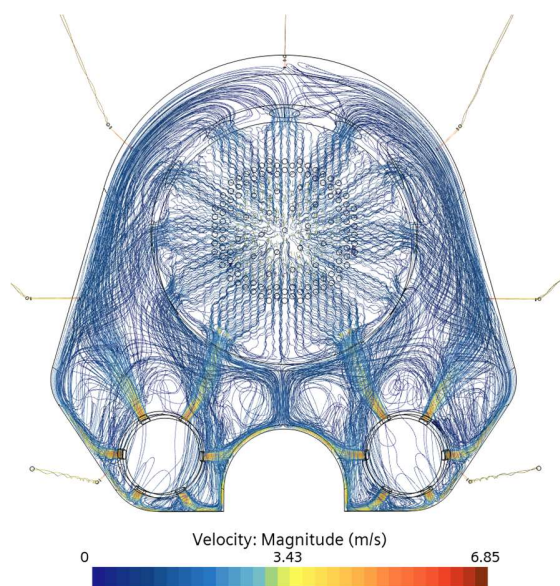


Figure 4. Streamlines in the lower plenum region, also showing the preliminary integration of pump heads into the region.

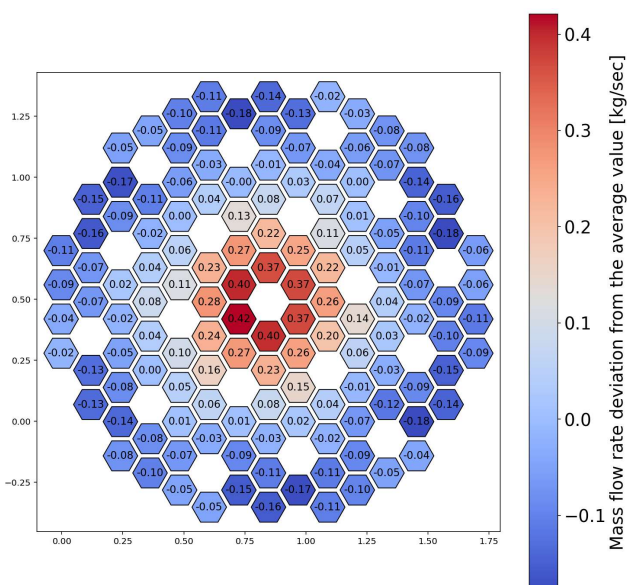


Figure 5. Deviation of the mass flow rate from the average value in the active core region in nominal condition.

Table III. Standard deviation on the core inlet mass flow rate, from the innermost to the outermost ring.

Core ring	1	2	3	4	5	6	7
Std (kg/s) – 2/ 2 pumps	0.022	0.025	0.040	0.034	0.040	0.038	0.030
Std (kg/s) – 1/ 2 pumps	0.035	0.027	0.026	0.022	0.022	0.021	0.019

Once the mechanical design of the lower plenum is finalized, further studies will be required to assess the reliability and robustness of the predicted flow distribution under a range of operating conditions. Special attention should be given to transient regimes involving transitions from nominal operation to hot shutdown and vice versa, including low mass flow rate conditions under both natural and forced convection. Such phases are crucial for maintenance and refueling activities and for accident scenarios, in which effective removal of residual core power critically depends on the lower plenum flow distribution.

3.2. System Thermal-hydraulics Analysis of the HEXANA Reactor

In the framework of the thermal-hydraulic analysis of the HEXANA reactor, a global dynamic model has been developed using the Dymola simulation environment [6]. The model includes the main components of both primary and intermediate sodium circuits, up to the sodium-molten salts heat exchanger. All subcomponent models representing the hydraulic circuits were developed or adapted from the Modelica Standard Library and the TRANSFORM library [7]. A point kinetics model has been implemented to describe the core neutronics. Global core reactivity feedback coefficients were considered (power coefficient $\alpha(P) = -0.4$ pcm/%P; core heating coefficient $\alpha(\Delta T) = -1.3$ pcm/°C; core inlet temperature coefficient $\alpha(T_{in}) = -2.5$ pcm/°C) [8]. Figure 6 shows the schematic layout of the model. Once the model reached the target steady-state conditions at 3600 seconds, it was used as the initial condition for simulating a set of representative operational transients. These include Unprotected Loss

Of Flow (ULOF) events, in which pump failures in the various circuits are modeled through ramp sources that progressively reduce the mass flow rate. In addition, Unprotected Loss Of Heat Sink (ULOHS) scenarios are considered by simulating a rapid degradation of heat transfer in the heat exchangers, representative of rapid emergency draining of the secondary or tertiary circuits. Given the fast time scale of this phenomenon, heat conduction through the exchanger walls is instantaneously disabled via a boolean input, representing the sudden transition to adiabatic conditions.

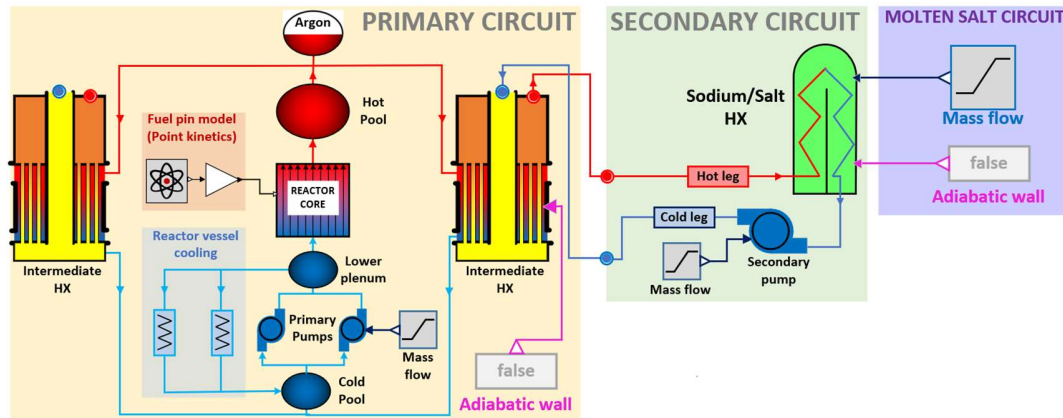


Figure 6. Schematic view of the HEXANA Reactor model in Dymola.

Figure 7 shows the system transient following the failure of one of the two secondary pumps, modeled as a linear reduction of the mass flow rate from nominal conditions to 5% over 10 seconds, representative of the hydraulic inertia associated with a pump trip. The resulting decrease in heat transfer within the intermediate heat exchanger leads to an increase in the primary-side core inlet temperature and a simultaneous reduction in core power and temperature jump. These effects introduce competing reactivity feedbacks: the inlet temperature rise produces negative reactivity, while the reduced power and core heating introduce positive reactivity. The transient evolves toward a new equilibrium state when these opposing contributions balance each other, after approximately 500 seconds.

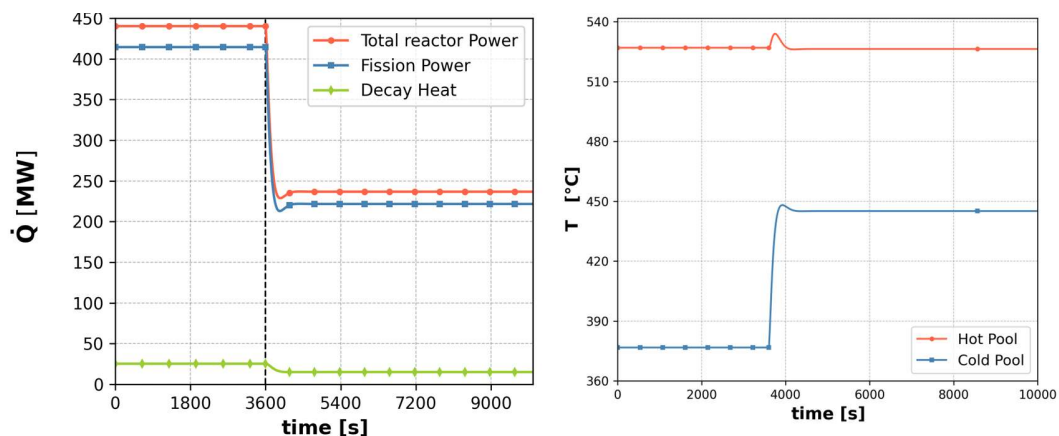


Figure 7. Reactor power and plena temperatures evolution during a ULOF scenario on the secondary circuit.

3.3. Neutronics Studies

To model the core and evaluate its performance, the HEXANA neutronics team developed a dedicated tool to generate the core datasets. This tool is based on the open-source Monte Carlo code OpenMC [9],

which enables both neutron transport and fuel depletion calculations during irradiation. It has been used in the “eigenvalue” mode. Nuclear data are taken from the JEFF-3.3 library. The simulation parameters are set to achieve a statistical precision of 20 pcm on reactivity during depletion calculations (5 pcm in static) and below 1% on the linear power. The chain used for the fuel isotopic depletion in the simulation is the complete one with 3820 isotopes.

Additional functionalities were implemented to account for HEXANA-specific features, such as control-rod operation strategies and core reload schemes. Leveraging the automation and Python scripting capabilities of the calculation tools, a one-third core reloading strategy was adopted. This means that one third of the assemblies are replaced at each cycle. The cycle length is determined according to a cladding damage criterion: in fast reactors, cladding mechanical endurance is generally more limiting than fuel burnup. Irradiation-induced swelling of the cladding must be restricted because excessive swelling degrades material’s ductility [10]. When volumetric swelling exceeds about 6%, the loss of ductility becomes unacceptable, and this threshold ultimately limits the allowable fast reactor fuel cycle length. The HEXANA core requires refueling every 14 months to maintain structural integrity and compliance with safety limits.

The reference HEXANA core achieves an effective multiplication factor of 1.05600 (rod extracted) at beginning of life (BOL), providing sufficient excess reactivity to cover availability and control requirements over the operating cycle. The core average linear power is 247 W/cm, and with control rods at their critical position, it varies between 155 W/cm to 348 W/cm, with a maximum-to-average assembly power peaking factor below 1.4, ensuring a homogeneous power distribution and stable thermal margins. Future studies will determine whether this value is sufficiently low or if it will need to be improved. Figure 8 shows the assembly-wise power distribution in the HEXANA core.

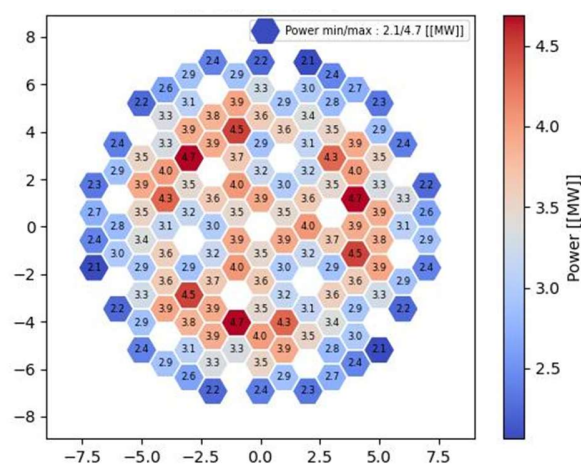


Figure 8. Fuel assembly power distribution at BOL at the assembly level.

The design optimization of enrichment zoning and control rod positioning leads to an almost flat radial flux profile and minimizes the power tilt between inner and outer regions. Reactivity coefficients (Doppler and coolant void) remain negative throughout the cycle, contributing to the intrinsic stability and safety of the core. The sodium void worth is maintained below 1β (which is close to 370 pcm), in line with the design objectives for compact sodium-cooled fast reactors.

4. CONCLUSIONS

The HEXANA project aims to develop a Sodium-cooled Fast Reactor (SFR)–type Advanced Modular Reactor (AMR) to provide flexible, low-carbon energy for industrial decarbonization. Its compact

primary system introduces spatial constraints and possible thermal-hydraulic asymmetries, requiring detailed assessment. Neutronics and thermal-hydraulics studies performed during the conceptual design phase have supported key design choices. CFD analyses of the lower plenum demonstrated a nearly uniform flow at the core inlet despite asymmetric pump layout, while dynamic simulations in Dymola confirmed stable behavior during ULOF and ULOHS transients. Using an in-house tool based on OpenMC, the HEXANA core was optimized to achieve an effective multiplication factor of 1.05 at Beginning of Life (BOL), an average power density of 247 W/cm³, and a peaking factor below 1.3. Reactivity coefficients remain negative and the sodium void worth below 1 β , confirming the core's inherent stability and compliance with fast-reactor safety standards. These results support the robustness of the HEXANA concept, demonstrating its viability as a flexible and efficient solution for industrial heat and power applications.

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